PRASAD V. POTLURI SIDDHARTHA INSTITUTE OF TECHNOLOGY
KANURU, VIJAYAWADA
II B.Tech – I Sem CSE (AI&ML)
ARTIFICIAL INTELLIGENCE
Micro Syllabus

Offering Branches:	CSE (AI&ML)	**Year/Semester:**	II/I
Course Category:	Engineering Science	**Credits:**	3
Course Type:	Theory	**Lecture-Tutorial-Practical:**	3-0-0
Prerequisites :	Discrete Mathematical Structures, Probability and Statistics	**Continuous Internal Evaluation:**	30
		Semester End Examination:	70
		Total Marks:	100

COURSE OUTCOMES

Upon successful completion of the course, the student will be able to:

CO1	Describe the principal and techniques of Artificial Intelligence including agents, problem-solving, knowledge representation, planning, and learning to gain a comprehensive understanding of intelligent systems	L2
CO2	Use uninformed and heuristic search strategies, and logic-based knowledge representation techniques to solve AI problems in deterministic and adversarial environments.	L3
CO3	Apply planning algorithms and learning techniques such as decision trees, reinforcement learning, and statistical learning to build intelligent autonomous agents.	L3
CO4	Analyze the effectiveness and suitability of AI search methods, knowledge representation, planning, and learning techniques for solving complex and real-world decision-making problems.	L4

Course Content

UNIT-1	**Introduction**: • **Definition of AI:** -Acting humanly: The Turing Test approach, -Thinking humanly: The cognitive modeling approach, -Thinking rationally: The "laws of thought" approach, -Acting rationally: The rational agent approach. • **History of AI:** -The gestation of artificial intelligence (1943–1955) -The birth of artificial intelligence (1956) -Early enthusiasm, great expectations (1952–1969) -A dose of reality (1966–1973) -Knowledge-based systems: The key to power (1969–1979) -AI becomes an industry (1980–present) -The return of neural networks (1986–present) -AI adopts the scientific method (1987–present)	CO1

	-The emergence of intelligent agents (1995–present) -The availability of very large data sets (2001–present) • **Foundations of AI:** -Philosophy -Mathematics -Economics -Neuroscience -Psychology -Computer engineering -Control theory and cybernetics -Linguistics • **Applications of AI:** -The state of the art. **Intelligent Agents:** • **Agents and Environments:** • **Good Behavior: The concept of rationality** -Rationality -Omniscience, Learning, and Autonomy • **The Nature of Environments** -Specifying the task environment -Properties of task environments • **The Structure of Agents:** -Agent programs -Simple reflex agents -Model-based reflex agents -Goal-based agents -Utility-based agents -Learning agents -How the components of agent programs work. • **Problem Solving Agents** -Well-defined problems and solutions -Formulating problems	
UNIT-2	**Searching** : • **Searching for Solutions** -Infrastructure for search algorithms -Measuring problem-solving performance **Uninformed Search Strategies:** • **Breadth first search** • **Depth first search** **Informed (Heuristic) Search Strategies:** • **Hill climbing** -Local maxima -Local beam search • **A* Algorithm** -A* search: Minimizing the total estimated solution cost -Conditions for optimality: Admissibility and consistency -Optimality of A* **Game Playing- Adversial search**:	CO1,CO2, CO4

	• **Games** mini-max algorithm, optimal decisions in multiplayer games • **Alpha-Beta Pruning** -Move ordering	
UNIT-3	**Knowledge Representation:** • **Logical Agents:** -Knowledge-Based Agents -Logic • **Propositional Logic:** -Syntax and semantics -A simple knowledge base -A simple inference procedure **First-Order Logic:** • **Syntax And Semantics** -Models for first-order logic -Symbols and interpretations -Terms -Atomic sentences -Complex sentences -Quantifiers -Equality -An alternative semantics. **Inference In First-Order Logic:** • **Propositional Vs. First-Order Inference** -Inference rules for quantifiers -Reduction to propositional inference • **Unification And Lifting** -A first order inference rule -unification -Storage and retrieval • **Forward Chaining** -First order definite clause -A simple forward- chaining algorithm • **Backward Chaining** -A Backward -Chaining algorithm -Logic programming • **Resolution** -Conjunctive normal form for first –order logic -The resolution inference rule -Example proofs -Completeness of resolution -Equality	CO1,CO2, CO4
UNIT-4	**Planning:** • **Classical Planning:** -Definition Of Classical Planning -Example problems. • **Planning with state space search:** -Forward (progression) state-space search	CO1,CO3, CO4

	-Backward (regression) relevant-states search -Heuristics for planning. • **Planning Graphs:** -Planning graphs for heuristic estimation -The GRAPHPLAN algorithm -Termination of GRAPHPLAN. • **Analysis Of Planning Approaches** • **Hierarchical Planning:** -High-level actions -Searching for primitive solutions -Searching for abstract solutions. • **Multi Agent Planning:** -Planning with multiple simultaneous actions -Planning with multiple agents:Cooperation and coordination	
UNIT-5	**Learning:** • **Learning from Examples:** -Forms Of Learning • **Learning Decision Trees** -The decision tree representation - Expressiveness of decision trees -Inducing decision trees from examples -Choosing attributes tests -Generalizations and over fitting -Broadening the applicability of decision trees • **Evaluating And Choosing The Best Hypothesis** -Model selection: Complexity versus goodness of fit -From error rates to loss -Regularization • **The Theory Of Learning** -PAC learning example: Learning decision lists • **Regression And Classification With Linear Models** - Univariate linear regression -Multivariate linear regression - Linear classifiers with a hard threshold - Linear classification with logistic regression • **Explanation based learning** -Extracting general rules from examples -Improving efficiency • **Statistical Learning methods** **Reinforcement Learning** **-Passive Reinforcement Learning** **-Active Reinforcement Learning**	CO1,CO3, CO4

Learning Resources

Text Books
1. Artificial Intelligence: A Modern Approach, Stuart Russell and Peter Norvig, Third Edition, Prentice Hall.
Reference Books
1. A Classical Approach to Artificial Intelligence, M.C. Trivedi, 2019, Khanna Book Publishing. 2. Artificial Intelligence, Elaine Rich and Kevin Knight, Tata McGraw Hill. 3. Artificial Intelligence Saroj Kaushik, 2011,Cengage Learning India.
e- Resources & other digital material
1. https://nptel.ac.in/courses/106105077 2. https://nptel.ac.in/courses/106105078 3. https://nptel.ac.in/courses/106106126 4. https://www.coursera.org/learn/introduction-to-ai 5. https://www.tutorialspoint.com/artificial_intelligence/index.htm

Course Coordinators:

Module Coordinator:

Program Coordinator:

HOD